

Dual simplified:

minimize

$$\sum_{vw \in E} z_{vw} M_{vw}$$

s.t.

$$y_r = 0$$

$$y_s = 1$$

$$y_w - y_v + z_{vw} \geq 0$$

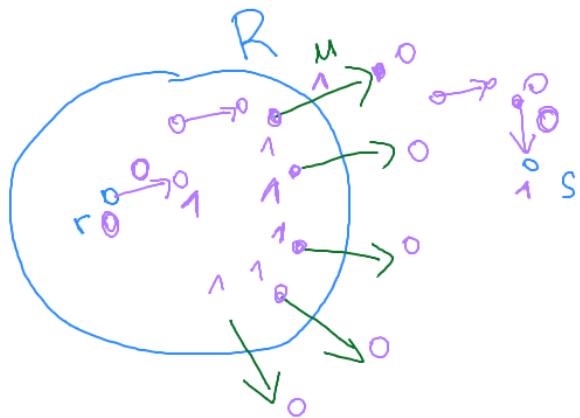
$$z_{vw} \geq 0$$

$$\forall vw \in E$$

$$\forall vw \in E$$

$$y_r = 0 \quad y_v \xrightarrow{z_{vw}} y_w \quad y_s = 1 \quad y_w \geq y_v - z_{vw}$$

for $\bar{\partial}(R)$ in rs-code



$$\min u(\bar{\partial}(R))$$

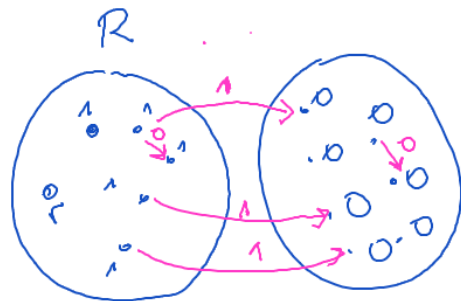
$$z_{vw} = \begin{cases} 1 & : vw \in \bar{\partial}(R) \\ 0 & : \text{cc} \end{cases}$$

$$y_u = \begin{cases} 1 & : u \in R \setminus r \\ 0 & : u \in \bar{R} \setminus s \end{cases}$$

$$u(R) \leq \text{sol}^* \leq \alpha(R)$$

Solvable binom (y, z) .

$$R = \{u : y_u = 1\} \cup \{r\}$$



$u(R)$ cost minimum = sol binom dual

Aplicación: Matching en bipartidos

Matching $M \subseteq E$ tq $e, f \in M \Rightarrow e \cap f = \emptyset$

Cobertura $K \subseteq V$ tq $uv \in E \Rightarrow \{u, v\} \cap K \neq \emptyset$

Prop 1 $|M| \leq |K| \quad \forall M, K$

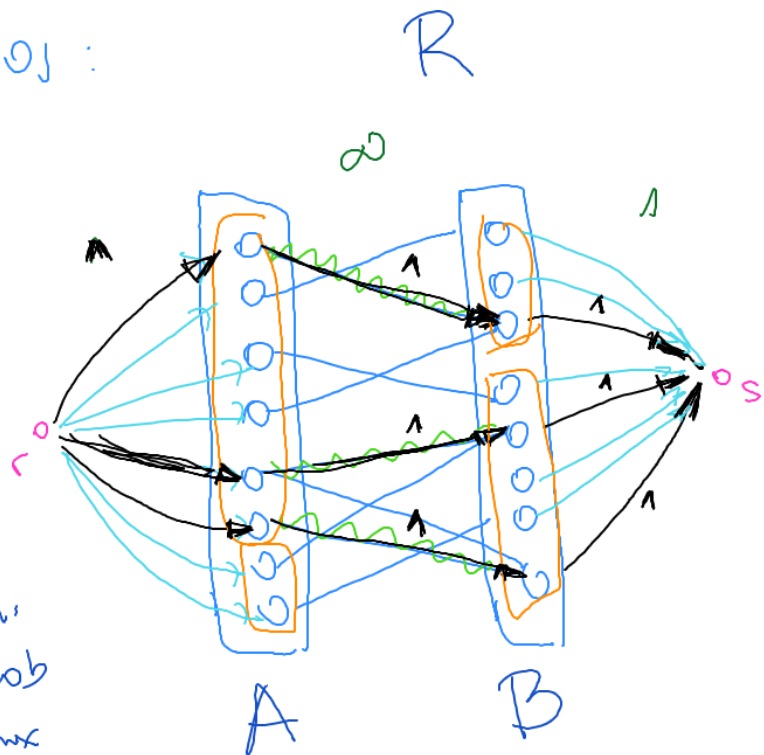
$\times q?$



Prop 2 $|M^*| = |K^*|$ bipartidos, (König) ✓



Conflyos :

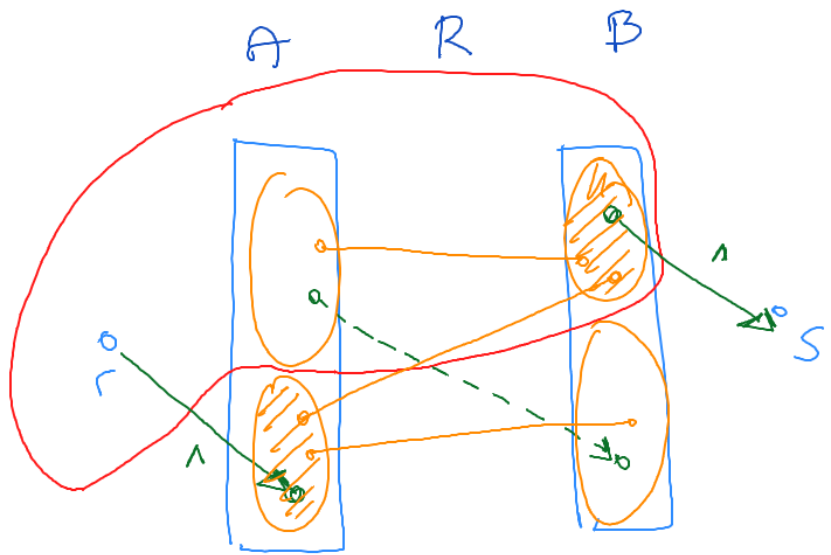


M^* : max min
 K^* : min cob
 x^* : flyomx
 R^* : wote mm

$$K = (A \setminus R) \cup (R \cap B)$$

$$|M^*| \stackrel{MPMC}{=} f(x^*) \stackrel{MPMC}{=} c(R^*) \stackrel{MPMC}{=} |K^*|$$

$$|M^*| \leq f(x^*)$$



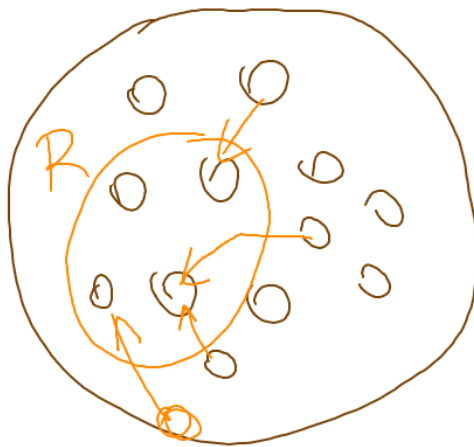
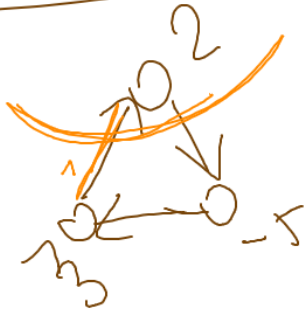
Problema de Gale

función demanda $b: V \rightarrow \mathbb{R}$, capacidades c

flujo x satisface b si $x^+(v) = b(v) \quad \forall v \in V$

Problema: encontrar x que satisfaga b y respete capacidades

Cond necesaria:



$$i) \quad b(V) = 0$$

$$\sum_u \frac{x^+(u)}{b(u)} = x(\partial^+(R)) - x(\partial^-(R)) \quad \checkmark$$

todo el mto $b(V) = x(\partial^+(V)) - x(\partial^-(V)) = 0$

$$ii) \quad b(X) \leq c^+(X) \quad \forall X \subseteq V$$

(∇ Gale: to en subconjuntos)